Generative Project

Milestone 2 — Proposal

Flickr30k Text-Conditioned Generation and Sampler Tuning

Group 9 — Team and Roles

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GitHub repository: https://github.com/Sneha0504/IE7615_Generative_Project

Objective and Model

Milestone 2 extends the Flickr30k text-embedding baseline from Milestone 1 into a verified text-conditioned image-generation workflow. Stable Diffusion 1.5 uses a frozen CLIP text encoder (77 × 768 embeddings), a latent U-Net (4 × 64 × 64 noise prediction with cross-attention), and a VAE decoder (512 × 512 RGB output). We implement a manual denoising loop, verify it against the Diffusers pipeline, and tune classifier-free guidance (CFG) and sampling schedules. No model weights are trained in this milestone.

Dataset and Experimental Design

The notebook reuses Milestone 1 processed Flickr30k images, caption splits, and saved embeddings (3,000 image–caption pairs; 2,400 / 300 / 300 train / validation / test). Four validation captions (V1–V4) are used for parameter selection; eight held-out test captions (T1–T8) are used for baseline and tuned generation. The baseline is DDIM with 30 steps and CFG 7.5. The independent CLIP image–text scorer reports 100 × cosine similarity; settings within 0.5 CLIP points of the best validation mean are treated as comparable, favoring lower guidance and fewer steps.

Executed Environment and Tuning Results

The notebook ran on Windows 11, Python 3.13.7, CPU-only PyTorch 2.6.0, Diffusers 0.35.1, and Transformers 4.49.0. The manual loop exactly matched the pipeline (maximum latent difference 0). Guidance and sampler sweeps selected CFG 15 and DDIM with 15 steps. All 8/8 validation checks passed. Across 68 generated images, the safety checker flagged one image.

Configuration	Validation CLIP	Mean time / image
DDIM 30 steps, CFG 15	34.96	95.91 s
DDIM 15 steps, CFG 15	34.76	61.36 s
DDIM 30 trailing, CFG 15	33.24	117.26 s
DPM-Solver++ 15, CFG 15	33.12	63.14 s

First Test Images: Baseline vs. Tuned

Metric	Baseline (DDIM-30, CFG 7.5)	Tuned (DDIM-15, CFG 15)
Mean CLIP score	35.46	36.31 (7 unflagged images)
Mean generation time	104.84 s	55.63 s
Image output	512 × 512 RGB	512 × 512 RGB

Observations and Next Steps

Higher guidance improved caption matching but sometimes oversaturated colors or misplaced attributes; the chosen CFG 15 should not be interpreted as best visual realism. DDIM with 15 steps nearly matched the 30-step validation CLIP score at substantially lower cost. On test captions, three tuned outputs scored higher and four lower than their baseline counterparts; one tuned image was safety-filtered. Some generated images omit context or mishandle relationships and negation. CLIP score measures text–image alignment, not photographic realism. Future work will assess realism (e.g., KID/FID) and explore LoRA fine-tuning on the 2,400 training pairs.